

10 分割の制約式

$$0 \leq Z_{10} + U_1 - 2 * W_{10} \leq 1$$

$$0 \leq Z_{20} + U_1 - 2 * W_{20} \leq 1$$

$$0 \leq Z_{30} + U_1 - 2 * W_{30} \leq 1$$

$$0 \leq Z_{40} + U_1 - 2 * W_{40} \leq 1$$

$$0 \leq Z_{50} + U_1 - 2 * W_{50} \leq 1$$

$$0 \leq Z_{60} + U_1 - 2 * W_{60} \leq 1$$

$$0 \leq Z_{70} + U_1 - 2 * W_{70} \leq 1$$

$$0 \leq Z_{80} + U_1 - 2 * W_{80} \leq 1$$

$$0 \leq Z_{90} + U_1 - 2 * W_{90} \leq 1$$

$$L_1 W_{10} \leq x_{11} \leq L_1 U_1$$

$$L_2 W_{20} \leq x_{21} \leq L_2 W_{10}$$

$$L_3 W_{30} \leq x_{31} \leq L_3 W_{20}$$

$$L_4 W_{40} \leq x_{41} \leq L_4 W_{30}$$

$$L_5 W_{50} \leq x_{51} \leq L_5 W_{40}$$

$$L_6 W_{60} \leq x_{61} \leq L_6 W_{50}$$

$$L_7 W_{70} \leq x_{71} \leq L_7 W_{60}$$

$$L_8 W_{80} \leq x_{81} \leq L_8 W_{70}$$

$$L_9 W_{90} \leq x_{91} \leq L_9 W_{80}$$

$$0 \leq x_{101} \leq L_{10} W_{90}$$

$$0 \leq Z_{10} + U_2 - 2 * W_{11} \leq 1$$

$$0 \leq Z_{20} + U_2 - 2 * W_{21} \leq 1$$

$$0 \leq Z_{30} + U_2 - 2 * W_{31} \leq 1$$

$$0 \leq Z_{40} + U_2 - 2 * W_{41} \leq 1$$

$$0 \leq Z_{50} + U_2 - 2 * W_{51} \leq 1$$

$$0 \leq Z_{60} + U_2 - 2 * W_{61} \leq 1$$

$$0 \leq Z_{70} + U_2 - 2 * W_{71} \leq 1$$

$$0 \leq Z_{80} + U_2 - 2 * W_{81} \leq 1$$

$$0 \leq Z_{90} + U_2 - 2 * W_{91} \leq 1$$

$$L_1 W_{11} \leq x_{12} \leq L_1 U_2$$

$$L_2 W_{21} \leq x_{22} \leq L_2 W_{11}$$

$$L_3 W_{31} \leq x_{32} \leq L_3 W_{21}$$

$$L_4 W_{41} \leq x_{42} \leq L_4 W_{31}$$

$$L_5 W_{51} \leq x_{52} \leq L_5 W_{41}$$

$$L_6 W_{61} \leq x_{62} \leq L_6 W_{51}$$

$$L_7 W_{71} \leq x_{72} \leq L_7 W_{61}$$

$$L_8 W_{81} \leq x_{82} \leq L_8 W_{71}$$

$$L_9 W_{91} \leq x_{92} \leq L_9 W_{81}$$

$$0 \leq x_{102} \leq L_{10} W_{91}$$

$$0 \leq Z_{10} + U_3 - 2 * W_{12} \leq 1$$

$$0 \leq Z_{20} + U_3 - 2 * W_{22} \leq 1$$

$$0 \leq Z_{30} + U_3 - 2 * W_{32} \leq 1$$

$$0 \leq Z_{40} + U_3 - 2 * W_{42} \leq 1$$

$$0 \leq Z_{50} + U_3 - 2 * W_{52} \leq 1$$

$$0 \leq Z_{60} + U_3 - 2 * W_{62} \leq 1$$

$$0 \leq Z_{70} + U_3 - 2 * W_{72} \leq 1$$

$$0 \leq Z_{80} + U_3 - 2 * W_{82} \leq 1$$

$$0 \leq Z_{90} + U_3 - 2 * W_{92} \leq 1$$

$$L_1 W_{12} \leq x_{13} \leq L_1 U_3$$

$$L_2 W_{22} \leq x_{23} \leq L_2 W_{12}$$

$$L_3 W_{32} \leq x_{33} \leq L_3 W_{22}$$

$$L_4 W_{42} \leq x_{43} \leq L_4 W_{32}$$

$$L_5 W_{52} \leq x_{53} \leq L_5 W_{42}$$

$$L_6 W_{62} \leq x_{63} \leq L_6 W_{52}$$

$$L_7 W_{72} \leq x_{73} \leq L_7 W_{62}$$

$$L_8 W_{82} \leq x_{83} \leq L_8 W_{72}$$

$$L_9 W_{92} \leq x_{93} \leq L_9 W_{82}$$

$$0 \leq x_{103} \leq L_{10} W_{92}$$

$$0 \leq Z_{10} + U_4 - 2 * W_{13} \leq 1$$

$$0 \leq Z_{20} + U_4 - 2 * W_{23} \leq 1$$

$$0 \leq Z_{30} + U_4 - 2 * W_{33} \leq 1$$

$$0 \leq Z_{40} + U_4 - 2 * W_{43} \leq 1$$

$$0 \leq Z_{50} + U_4 - 2 * W_{53} \leq 1$$

$$0 \leq Z_{60} + U_4 - 2 * W_{63} \leq 1$$

$$0 \leq Z_{70} + U_4 - 2 * W_{73} \leq 1$$

$$0 \leq Z_{80} + U_4 - 2 * W_{83} \leq 1$$

$$0 \leq Z_{90} + U_4 - 2 * W_{93} \leq 1$$

$$L_1 W_{13} \leq x_{14} \leq L_1 U_4$$

$$L_2 W_{23} \leq x_{24} \leq L_2 W_{13}$$

$$L_3 W_{33} \leq x_{34} \leq L_3 W_{23}$$

$$L_4 W_{43} \leq x_{44} \leq L_4 W_{33}$$

$$L_5 W_{53} \leq x_{54} \leq L_5 W_{43}$$

$$L_6 W_{63} \leq x_{64} \leq L_6 W_{53}$$

$$L_7 W_{73} \leq x_{74} \leq L_7 W_{63}$$

$$L_8 W_{83} \leq x_{84} \leq L_8 W_{73}$$

$$L_9 W_{93} \leq x_{94} \leq L_9 W_{83}$$

$$0 \leq x_{104} \leq L_{10} W_{93}$$

$$0 \leq Z_{10} + U_5 - 2 * W_{14} \leq 1$$

$$0 \leq Z_{20} + U_5 - 2 * W_{24} \leq 1$$

$$0 \leq Z_{30} + U_5 - 2 * W_{34} \leq 1$$

$$0 \leq Z_{40} + U_5 - 2 * W_{44} \leq 1$$

$$0 \leq Z_{50} + U_5 - 2 * W_{54} \leq 1$$

$$0 \leq Z_{60} + U_5 - 2 * W_{64} \leq 1$$

$$0 \leq Z_{70} + U_5 - 2 * W_{74} \leq 1$$

$$0 \leq Z_{80} + U_5 - 2 * W_{84} \leq 1$$

$$0 \leq Z_{90} + U_5 - 2 * W_{94} \leq 1$$

$$L_1 W_{14} \leq x_{15} \leq L_1 U_5$$

$$L_2 W_{24} \leq x_{25} \leq L_2 W_{14}$$

$$L_3 W_{34} \leq x_{35} \leq L_3 W_{24}$$

$$L_4 W_{44} \leq x_{45} \leq L_4 W_{34}$$

$$L_5 W_{54} \leq x_{55} \leq L_5 W_{44}$$

$$L_6 W_{64} \leq x_{65} \leq L_6 W_{54}$$

$$L_7 W_{74} \leq x_{75} \leq L_7 W_{64}$$

$$L_8 W_{84} \leq x_{85} \leq L_8 W_{74}$$

$$L_9 W_{94} \leq x_{95} \leq L_9 W_{84}$$

$$0 \leq x_{105} \leq L_{10} W_{94}$$

$$0 \leq Z_{10} + U_6 - 2 * W_{15} \leq 1$$

$$0 \leq Z_{20} + U_6 - 2 * W_{25} \leq 1$$

$$0 \leq Z_{30} + U_6 - 2 * W_{35} \leq 1$$

$$0 \leq Z_{40} + U_6 - 2 * W_{45} \leq 1$$

$$0 \leq Z_{50} + U_6 - 2 * W_{55} \leq 1$$

$$0 \leq Z_{60} + U_6 - 2 * W_{65} \leq 1$$

$$0 \leq Z_{70} + U_6 - 2 * W_{75} \leq 1$$

$$0 \leq Z_{80} + U_6 - 2 * W_{85} \leq 1$$

$$0 \leq Z_{90} + U_6 - 2 * W_{95} \leq 1$$

$$L_1 W_{15} \leq x_{16} \leq L_1 U_6$$

$$L_2 W_{25} \leq x_{26} \leq L_2 W_{15}$$

$$L_3 W_{35} \leq x_{36} \leq L_3 W_{25}$$

$$L_4 W_{45} \leq x_{46} \leq L_4 W_{35}$$

$$L_5 W_{55} \leq x_{56} \leq L_5 W_{45}$$

$$L_6 W_{65} \leq x_{66} \leq L_6 W_{55}$$

$$L_7 W_{75} \leq x_{76} \leq L_7 W_{65}$$

$$L_8 W_{85} \leq x_{86} \leq L_8 W_{75}$$

$$L_9 W_{95} \leq x_{96} \leq L_9 W_{85}$$

$$0 \leq x_{106} \leq L_{10} W_{95}$$

$$0 \leq Z_{10} + U_7 - 2 * W_{16} \leq 1$$

$$0 \leq Z_{20} + U_7 - 2 * W_{26} \leq 1$$

$$0 \leq Z_{30} + U_7 - 2 * W_{36} \leq 1$$

$$0 \leq Z_{40} + U_7 - 2 * W_{46} \leq 1$$

$$0 \leq Z_{50} + U_7 - 2 * W_{56} \leq 1$$

$$0 \leq Z_{60} + U_7 - 2 * W_{66} \leq 1$$

$$0 \leq Z_{70} + U_7 - 2 * W_{76} \leq 1$$

$$0 \leq Z_{80} + U_7 - 2 * W_{86} \leq 1$$

$$0 \leq Z_{90} + U_7 - 2 * W_{96} \leq 1$$

$$L_1 W_{16} \leq x_{17} \leq L_1 U_7$$

$$L_2W_{26} \leq x_{27} \leq L_2W_{16}$$

$$L_3W_{36} \leq x_{37} \leq L_3W_{26}$$

$$L_4W_{46} \leq x_{47} \leq L_4W_{36}$$

$$L_5W_{56} \leq x_{57} \leq L_5W_{46}$$

$$L_6W_{66} \leq x_{67} \leq L_6W_{56}$$

$$L_7W_{76} \leq x_{77} \leq L_7W_{66}$$

$$L_8W_{86} \leq x_{87} \leq L_8W_{76}$$

$$L_9W_{96} \leq x_{97} \leq L_9W_{86}$$

$$0 \leq x_{107} \leq L_{10}W_{96}$$

$$0 \leq Z_{10} + U_8 - 2 * W_{17} \leq 1$$

$$0 \leq Z_{20} + U_8 - 2 * W_{27} \leq 1$$

$$0 \leq Z_{30} + U_8 - 2 * W_{37} \leq 1$$

$$0 \leq Z_{40} + U_8 - 2 * W_{47} \leq 1$$

$$0 \leq Z_{50} + U_8 - 2 * W_{57} \leq 1$$

$$0 \leq Z_{60} + U_8 - 2 * W_{67} \leq 1$$

$$0 \leq Z_{70} + U_8 - 2 * W_{77} \leq 1$$

$$0 \leq Z_{80} + U_8 - 2 * W_{87} \leq 1$$

$$0 \leq Z_{90} + U_8 - 2 * W_{97} \leq 1$$

$$L_1W_{17} \leq x_{18} \leq L_1U_8$$

$$L_2W_{27} \leq x_{28} \leq L_2W_{17}$$

$$L_3W_{37} \leq x_{38} \leq L_3W_{27}$$

$$L_4W_{47} \leq x_{48} \leq L_4W_{37}$$

$$L_5W_{57} \leq x_{58} \leq L_5W_{47}$$

$$L_6W_{67} \leq x_{68} \leq L_6W_{57}$$

$$L_7W_{77} \leq x_{78} \leq L_7W_{67}$$

$$L_8W_{87} \leq x_{88} \leq L_8W_{77}$$

$$L_9W_{97} \leq x_{98} \leq L_9W_{87}$$

$$0 \leq x_{108} \leq L_{10}W_{97}$$

$$0 \leq Z_{10} + U_9 - 2 * W_{18} \leq 1$$

$$0 \leq Z_{20} + U_9 - 2 * W_{28} \leq 1$$

$$0 \leq Z_{30} + U_9 - 2 * W_{38} \leq 1$$

$$0 \leq Z_{40} + U_9 - 2 * W_{48} \leq 1$$

$$0 \leq Z_{50} + U_9 - 2 * W_{58} \leq 1$$

$$0 \leq Z_{60} + U_9 - 2 * W_{68} \leq 1$$

$$0 \leq Z_{70} + U_9 - 2 * W_{78} \leq 1$$

$$0 \leq Z_{80} + U_9 - 2 * W_{88} \leq 1$$

$$0 \leq Z_{90} + U_9 - 2 * W_{98} \leq 1$$

$$L_1W_{18} \leq x_{19} \leq L_1U_9$$

$$L_2W_{28} \leq x_{29} \leq L_2W_{18}$$

$$L_3W_{38} \leq x_{39} \leq L_3W_{28}$$

$$L_4W_{48} \leq x_{49} \leq L_4W_{38}$$

$$L_5W_{58} \leq x_{59} \leq L_5W_{48}$$

$$L_6W_{68} \leq x_{69} \leq L_6W_{58}$$

$$L_7W_{78} \leq x_{79} \leq L_7W_{68}$$

$$L_8W_{88} \leq x_{89} \leq L_8W_{78}$$

$$L_9W_{98} \leq x_{99} \leq L_9W_{88}$$

$$0 \leq x_{109} \leq L_{10}W_{98}$$

$$0 \leq Z_{10} + U_{10} - 2 * W_{19} \leq 1$$

$$0 \leq Z_{20} + U_{10} - 2 * W_{29} \leq 1$$

$$0 \leq Z_{30} + U_{10} - 2 * W_{39} \leq 1$$

$$0 \leq Z_{40} + U_{10} - 2 * W_{49} \leq 1$$

$$0 \leq Z_{50} + U_{10} - 2 * W_{59} \leq 1$$

$$0 \leq Z_{60} + U_{10} - 2 * W_{69} \leq 1$$

$$0 \leq Z_{70} + U_{10} - 2 * W_{79} \leq 1$$

$$0 \leq Z_{80} + U_{10} - 2 * W_{89} \leq 1$$

$$0 \leq Z_{90} + U_{10} - 2 * W_{99} \leq 1$$

$$L_1W_{19} \leq x_{110} \leq L_1U_{10}$$

$$L_2W_{29} \leq x_{210} \leq L_2W_{19}$$

$$L_3W_{39} \leq x_{310} \leq L_3W_{29}$$

$$L_4W_{49} \leq x_{410} \leq L_4W_{39}$$

$$L_5W_{59} \leq x_{510} \leq L_5W_{49}$$

$$L_6W_{69} \leq x_{610} \leq L_6W_{59}$$

$$L_7W_{79} \leq x_{710} \leq L_7W_{69}$$

$$L_8 W_{89} \leq x_{810} \leq L_8 W_{79}$$

$$L_9 W_{99} \leq x_{910} \leq L_9 W_{89}$$

$$0 \leq x_{1010} \leq L_{10} W_{99}$$

$$Z_{10} = 1 - Z_{10}$$

$$Z_{20} = Z_{10}(1 - Z_{20})$$

$$Z_{30} = Z_{20}(1 - Z_{30})$$

$$Z_{40} = Z_{30}(1 - Z_{40})$$

$$Z_{50} = Z_{40}(1 - Z_{50})$$

$$Z_{60} = Z_{50}(1 - Z_{60})$$

$$Z_{70} = Z_{60}(1 - Z_{70})$$

$$Z_{80} = Z_{70}(1 - Z_{80})$$

$$Z_{90} = Z_{80}(1 - Z_{90})$$

$$Z_{90}$$

$$0 \leq Z'_{10} + U_1 - 2 * A_{11} \leq 1$$

$$0 \leq Z'_{10} + U_2 - 2 * A_{12} \leq 1$$

$$0 \leq Z'_{10} + U_3 - 2 * A_{13} \leq 1$$

$$0 \leq Z'_{10} + U_4 - 2 * A_{14} \leq 1$$

$$0 \leq Z'_{10} + U_5 - 2 * A_{15} \leq 1$$

$$0 \leq Z'_{10} + U_6 - 2 * A_{16} \leq 1$$

$$0 \leq Z'_{10} + U_7 - 2 * A_{17} \leq 1$$

$$0 \leq Z'_{10} + U_8 - 2 * A_{18} \leq 1$$

$$0 \leq Z'_{10} + U_9 - 2 * A_{19} \leq 1$$

$$0 \leq Z'_{10} + U_{10} - 2 * A_{110} \leq 1$$

$$\begin{aligned} L_1(A_{12} + A_{13} + A_{14} + A_{15} + A_{16} + A_{17} + A_{18} + A_{19} + A_{110}) &\leq y_{11} \\ &\leq L_1(A_{11} + A_{12} + A_{13} + A_{14} + A_{15} + A_{16} + A_{17} + A_{18} + A_{19} + A_{110}) \end{aligned}$$

$$\begin{aligned} L_2(A_{13} + A_{14} + A_{15} + A_{16} + A_{17} + A_{18} + A_{19} + A_{110}) &\leq y_{12} \\ &\leq L_2(A_{12} + A_{13} + A_{14} + A_{15} + A_{16} + A_{17} + A_{18} + A_{19} + A_{110}) \end{aligned}$$

$$\begin{aligned} L_3(A_{14} + A_{15} + A_{16} + A_{17} + A_{18} + A_{19} + A_{110}) &\leq y_{13} \\ &\leq L_3(A_{13} + A_{14} + A_{15} + A_{16} + A_{17} + A_{18} + A_{19} + A_{110}) \end{aligned}$$

$$L_4(A_{15} + A_{16} + A_{17} + A_{18} + A_{19} + A_{110}) \leq y_{14} \leq L_4(A_{14} + A_{15} + A_{16} + A_{17} + A_{18} + A_{19} + A_{110})$$

$$L_5(A_{16} + A_{17} + A_{18} + A_{19} + A_{110}) \leq y_{15} \leq L_5(A_{15} + A_{16} + A_{17} + A_{18} + A_{19} + A_{110})$$

$$L_6(A_{17} + A_{18} + A_{19} + A_{110}) \leq y_{16} \leq L_6(A_{16} + A_{17} + A_{18} + A_{19} + A_{110})$$

$$L_6(A_{18} + A_{19} + A_{110}) \leq y_{17} \leq L_7(A_{17} + A_{18} + A_{19} + A_{110})$$

$$L_8(A_{19} + A_{110}) \leq y_{18} \leq L_8(A_{18} + A_{19} + A_{110})$$

$$L_9(A_{110}) \leq y_{19} \leq L_9(A_{19} + A_{110})$$

$$0 \leq y_{110} \leq L_{10}(A_{110})$$

$$0 \leq Z'_{20} + U_1 - 2 * A_{21} \leq 1$$

$$0 \leq Z'_{20} + U_2 - 2 * A_{22} \leq 1$$

$$0 \leq Z'_{20} + U_3 - 2 * A_{23} \leq 1$$

$$0 \leq Z'_{20} + U_4 - 2 * A_{24} \leq 1$$

$$0 \leq Z'_{20} + U_5 - 2 * A_{25} \leq 1$$

$$0 \leq Z'_{20} + U_6 - 2 * A_{26} \leq 1$$

$$0 \leq Z'_{20} + U_7 - 2 * A_{27} \leq 1$$

$$0 \leq Z'_{20} + U_8 - 2 * A_{28} \leq 1$$

$$0 \leq Z'_{20} + U_9 - 2 * A_{29} \leq 1$$

$$0 \leq Z'_{20} + U_{10} - 2 * A_{210} \leq 1$$

$$\begin{aligned} L_1(A_{22} + A_{23} + A_{24} + A_{25} + A_{26} + A_{27} + A_{28} + A_{29} + A_{210}) &\leq y_{21} \\ &\leq L_1(A_{21} + A_{22} + A_{23} + A_{24} + A_{25} + A_{26} + A_{27} + A_{28} + A_{29} + A_{210}) \end{aligned}$$

$$\begin{aligned} L_2(A_{23} + A_{24} + A_{25} + A_{26} + A_{27} + A_{28} + A_{29} + A_{210}) &\leq y_{22} \\ &\leq L_2(A_{22} + A_{23} + A_{24} + A_{25} + A_{26} + A_{27} + A_{28} + A_{29} + A_{210}) \end{aligned}$$

$$\begin{aligned} L_3(A_{24} + A_{25} + A_{26} + A_{27} + A_{28} + A_{29} + A_{210}) &\leq y_{23} \\ &\leq L_3(A_{23} + A_{24} + A_{25} + A_{26} + A_{27} + A_{28} + A_{29} + A_{210}) \end{aligned}$$

$$\begin{aligned} L_4(A_{25} + A_{26} + A_{27} + A_{28} + A_{29} + A_{210}) &\leq y_{24} \\ &\leq L_4(A_{24} + A_{25} + A_{26} + A_{27} + A_{28} + A_{29} + A_{210}) \end{aligned}$$

$$L_5(A_{26} + A_{27} + A_{28} + A_{29} + A_{210}) \leq y_{25} \leq L_5(A_{25} + A_{26} + A_{27} + A_{28} + A_{29} + A_{210})$$

$$L_6(A_{27} + A_{28} + A_{29} + A_{210}) \leq y_{26} \leq L_6(A_{26} + A_{27} + A_{28} + A_{29} + A_{210})$$

$$L_6(A_{28} + A_{29} + A_{210}) \leq y_{27} \leq L_7(A_{27} + A_{28} + A_{29} + A_{210})$$

$$L_8(A_{29} + A_{210}) \leq y_{28} \leq L_8(A_{28} + A_{29} + A_{210})$$

$$L_9(A_{210}) \leq y_{29} \leq L_9(A_{29} + A_{210})$$

$$0 \leq y_{210} \leq L_{10}(A_{210})$$

$$0 \leq Z'_{30} + U_1 - 2 * A_{31} \leq 1$$

$$0 \leq Z'_{30} + U_2 - 2 * A_{32} \leq 1$$

$$0 \leq Z'_{30} + U_3 - 2 * A_{33} \leq 1$$

$$0 \leq Z'_{30} + U_4 - 2 * A_{34} \leq 1$$

$$0 \leq Z'_{30} + U_5 - 2 * A_{35} \leq 1$$

$$0 \leq Z'_{30} + U_6 - 2 * A_{36} \leq 1$$

$$0 \leq Z'_{30} + U_7 - 2 * A_{37} \leq 1$$

$$0 \leq Z'_{30} + U_8 - 2 * A_{38} \leq 1$$

$$0 \leq Z'_{30} + U_9 - 2 * A_{39} \leq 1$$

$$0 \leq Z'_{30} + U_{10} - 2 * A_{310} \leq 1$$

$$\begin{aligned} L_1(A_{32} + A_{33} + A_{34} + A_{35} + A_{36} + A_{37} + A_{38} + A_{39} + A_{310}) &\leq y_{31} \\ &\leq L_1(A_{31} + A_{32} + A_{33} + A_{34} + A_{35} + A_{36} + A_{37} + A_{38} + A_{39} + A_{310}) \end{aligned}$$

$$\begin{aligned} L_2(A_{33} + A_{34} + A_{35} + A_{36} + A_{37} + A_{38} + A_{39} + A_{310}) &\leq y_{32} \\ &\leq L_2(A_{32} + A_{33} + A_{34} + A_{35} + A_{36} + A_{37} + A_{38} + A_{39} + A_{310}) \end{aligned}$$

$$\begin{aligned} L_3(A_{34} + A_{35} + A_{36} + A_{37} + A_{38} + A_{39} + A_{310}) &\leq y_{33} \\ &\leq L_3(A_{33} + A_{34} + A_{35} + A_{36} + A_{37} + A_{38} + A_{39} + A_{310}) \end{aligned}$$

$$\begin{aligned} L_4(A_{35} + A_{36} + A_{37} + A_{38} + A_{39} + A_{310}) &\leq y_{34} \\ &\leq L_4(A_{34} + A_{35} + A_{36} + A_{37} + A_{38} + A_{39} + A_{310}) \end{aligned}$$

$$L_5(A_{36} + A_{37} + A_{38} + A_{39} + A_{310}) \leq y_{35} \leq L_5(A_{35} + A_{36} + A_{37} + A_{38} + A_{39} + A_{310})$$

$$L_6(A_{37} + A_{38} + A_{39} + A_{310}) \leq y_{36} \leq L_6(A_{36} + A_{37} + A_{38} + A_{39} + A_{310})$$

$$L_6(A_{38} + A_{39} + A_{310}) \leq y_{37} \leq L_7(A_{37} + A_{38} + A_{39} + A_{310})$$

$$L_8(A_{39} + A_{310}) \leq y_{38} \leq L_8(A_{38} + A_{39} + A_{310})$$

$$L_9(A_{310}) \leq y_{39} \leq L_9(A_{39} + A_{310})$$

$$0 \leq y_{310} \leq L_{30}(A_{310})$$

$$0 \leq Z'_{40} + U_1 - 2 * A_{41} \leq 1$$

$$0 \leq Z'_{40} + U_2 - 2 * A_{42} \leq 1$$

$$0 \leq Z'_{40} + U_3 - 2 * A_{43} \leq 1$$

$$0 \leq Z'_{40} + U_4 - 2 * A_{44} \leq 1$$

$$0 \leq Z'_{40} + U_5 - 2 * A_{45} \leq 1$$

$$0 \leq Z'_{40} + U_6 - 2 * A_{46} \leq 1$$

$$0 \leq Z'_{40} + U_7 - 2 * A_{47} \leq 1$$

$$0 \leq Z'_{40} + U_8 - 2 * A_{48} \leq 1$$

$$0 \leq Z'_{40} + U_9 - 2 * A_{49} \leq 1$$

$$0 \leq Z'_{40} + U_{10} - 2 * A_{410} \leq 1$$

$$\begin{aligned} L_1(A_{42} + A_{43} + A_{44} + A_{45} + A_{46} + A_{47} + A_{48} + A_{49} + A_{410}) &\leq y_{41} \\ &\leq L_1(A_{41} + A_{42} + A_{43} + A_{44} + A_{45} + A_{46} + A_{47} + A_{48} + A_{49} + A_{410}) \end{aligned}$$

$$\begin{aligned} L_2(A_{43} + A_{44} + A_{45} + A_{46} + A_{47} + A_{48} + A_{49} + A_{410}) &\leq y_{42} \\ &\leq L_2(A_{42} + A_{43} + A_{44} + A_{45} + A_{46} + A_{47} + A_{48} + A_{49} + A_{410}) \end{aligned}$$

$$\begin{aligned} L_3(A_{44} + A_{45} + A_{46} + A_{47} + A_{48} + A_{49} + A_{410}) &\leq y_{43} \\ &\leq L_3(A_{43} + A_{44} + A_{45} + A_{46} + A_{47} + A_{48} + A_{49} + A_{410}) \end{aligned}$$

$$\begin{aligned} L_4(A_{45} + A_{46} + A_{47} + A_{48} + A_{49} + A_{410}) &\leq y_{44} \\ &\leq L_4(A_{44} + A_{45} + A_{46} + A_{47} + A_{48} + A_{49} + A_{410}) \end{aligned}$$

$$L_5(A_{46} + A_{47} + A_{48} + A_{49} + A_{410}) \leq y_{45} \leq L_5(A_{45} + A_{46} + A_{47} + A_{48} + A_{49} + A_{410})$$

$$L_6(A_{47} + A_{48} + A_{49} + A_{410}) \leq y_{46} \leq L_6(A_{46} + A_{47} + A_{48} + A_{49} + A_{410})$$

$$L_6(A_{48} + A_{49} + A_{410}) \leq y_{47} \leq L_7(A_{47} + A_{48} + A_{49} + A_{410})$$

$$L_8(A_{49} + A_{410}) \leq y_{48} \leq L_8(A_{48} + A_{49} + A_{410})$$

$$L_9(A_{410}) \leq y_{49} \leq L_9(A_{49} + A_{410})$$

$$0 \leq y_{410} \leq L_{10}(A_{410})$$

$$0 \leq Z'_{50} + U_1 - 2 * A_{51} \leq 1$$

$$0 \leq Z'_{50} + U_2 - 2 * A_{52} \leq 1$$

$$0 \leq Z'_{50} + U_3 - 2 * A_{53} \leq 1$$

$$0 \leq Z'_{50} + U_4 - 2 * A_{54} \leq 1$$

$$0 \leq Z'_{50} + U_5 - 2 * A_{55} \leq 1$$

$$0 \leq Z'_{50} + U_6 - 2 * A_{56} \leq 1$$

$$0 \leq Z'_{50} + U_7 - 2 * A_{57} \leq 1$$

$$0 \leq Z'_{50} + U_8 - 2 * A_{58} \leq 1$$

$$0 \leq Z'_{50} + U_9 - 2 * A_{59} \leq 1$$

$$0 \leq Z'_{50} + U_{10} - 2 * A_{510} \leq 1$$

$$\begin{aligned} L_1(A_{52} + A_{53} + A_{54} + A_{55} + A_{56} + A_{57} + A_{58} + A_{59} + A_{510}) &\leq y_{51} \\ &\leq L_1(A_{51} + A_{52} + A_{53} + A_{54} + A_{55} + A_{56} + A_{57} + A_{58} + A_{59} + A_{510}) \end{aligned}$$

$$\begin{aligned} L_2(A_{53} + A_{54} + A_{55} + A_{56} + A_{57} + A_{58} + A_{59} + A_{510}) &\leq y_{52} \\ &\leq L_2(A_{52} + A_{53} + A_{54} + A_{55} + A_{56} + A_{57} + A_{58} + A_{59} + A_{510}) \end{aligned}$$

$$\begin{aligned} L_3(A_{54} + A_{55} + A_{56} + A_{57} + A_{58} + A_{59} + A_{510}) &\leq y_{53} \\ &\leq L_3(A_{53} + A_{54} + A_{55} + A_{56} + A_{57} + A_{58} + A_{59} + A_{510}) \end{aligned}$$

$$\begin{aligned} L_4(A_{55} + A_{56} + A_{57} + A_{58} + A_{59} + A_{510}) &\leq y_{54} \\ &\leq L_4(A_{54} + A_{55} + A_{56} + A_{57} + A_{58} + A_{59} + A_{510}) \end{aligned}$$

$$L_5(A_{56} + A_{57} + A_{58} + A_{59} + A_{510}) \leq y_{55} \leq L_5(A_{55} + A_{56} + A_{57} + A_{58} + A_{59} + A_{510})$$

$$L_6(A_{57} + A_{58} + A_{59} + A_{510}) \leq y_{56} \leq L_6(A_{56} + A_{57} + A_{58} + A_{59} + A_{510})$$

$$L_6(A_{58} + A_{59} + A_{510}) \leq y_{57} \leq L_7(A_{57} + A_{58} + A_{59} + A_{510})$$

$$L_8(A_{59} + A_{510}) \leq y_{58} \leq L_8(A_{58} + A_{59} + A_{510})$$

$$L_9(A_{510}) \leq y_{59} \leq L_9(A_{59} + A_{510})$$

$$0 \leq y_{510} \leq L_{10}(A_{510})$$

$$0 \leq Z'_{60} + U_1 - 2 * A_{61} \leq 1$$

$$0 \leq Z'_{60} + U_2 - 2 * A_{62} \leq 1$$

$$0 \leq Z'_{60} + U_3 - 2 * A_{63} \leq 1$$

$$0 \leq Z'_{60} + U_4 - 2 * A_{64} \leq 1$$

$$0 \leq Z'_{60} + U_5 - 2 * A_{65} \leq 1$$

$$0 \leq Z'_{60} + U_6 - 2 * A_{66} \leq 1$$

$$0 \leq Z'_{60} + U_7 - 2 * A_{67} \leq 1$$

$$0 \leq Z'_{60} + U_8 - 2 * A_{68} \leq 1$$

$$0 \leq Z'_{60} + U_9 - 2 * A_{69} \leq 1$$

$$0 \leq Z'_{60} + U_{10} - 2 * A_{610} \leq 1$$

$$\begin{aligned} L_1(A_{62} + A_{63} + A_{64} + A_{65} + A_{66} + A_{67} + A_{68} + A_{69} + A_{610}) &\leq y_{61} \\ &\leq L_1(A_{61} + A_{62} + A_{63} + A_{64} + A_{65} + A_{66} + A_{67} + A_{68} + A_{69} + A_{610}) \end{aligned}$$

$$\begin{aligned} L_2(A_{63} + A_{64} + A_{65} + A_{66} + A_{67} + A_{68} + A_{69} + A_{610}) &\leq y_{62} \\ &\leq L_2(A_{62} + A_{63} + A_{64} + A_{65} + A_{66} + A_{67} + A_{68} + A_{69} + A_{610}) \end{aligned}$$

$$\begin{aligned} L_3(A_{64} + A_{65} + A_{66} + A_{67} + A_{68} + A_{69} + A_{610}) &\leq y_{63} \\ &\leq L_3(A_{63} + A_{64} + A_{65} + A_{66} + A_{67} + A_{68} + A_{69} + A_{610}) \end{aligned}$$

$$\begin{aligned}
L_4(A_{65} + A_{66} + A_{67} + A_{68} + A_{69} + A_{610}) &\leq y_{64} \\
&\leq L_4(A_{64} + A_{65} + A_{66} + A_{67} + A_{68} + A_{69} + A_{610})
\end{aligned}$$

$$L_5(A_{66} + A_{67} + A_{68} + A_{69} + A_{610}) \leq y_{65} \leq L_5(A_{65} + A_{66} + A_{67} + A_{68} + A_{69} + A_{610})$$

$$L_6(A_{67} + A_{68} + A_{69} + A_{610}) \leq y_{66} \leq L_6(A_{66} + A_{67} + A_{68} + A_{69} + A_{610})$$

$$L_6(A_{68} + A_{69} + A_{610}) \leq y_{67} \leq L_7(A_{67} + A_{68} + A_{69} + A_{610})$$

$$L_8(A_{69} + A_{610}) \leq y_{68} \leq L_8(A_{68} + A_{69} + A_{610})$$

$$L_9(A_{610}) \leq y_{69} \leq L_9(A_{69} + A_{610})$$

$$0 \leq y_{610} \leq L_{10}(A_{610})$$

$$0 \leq Z'_{70} + U_1 - 2 * A_{71} \leq 1$$

$$0 \leq Z'_{70} + U_2 - 2 * A_{72} \leq 1$$

$$0 \leq Z'_{70} + U_3 - 2 * A_{73} \leq 1$$

$$0 \leq Z'_{70} + U_4 - 2 * A_{74} \leq 1$$

$$0 \leq Z'_{70} + U_5 - 2 * A_{75} \leq 1$$

$$0 \leq Z'_{70} + U_6 - 2 * A_{76} \leq 1$$

$$0 \leq Z'_{70} + U_7 - 2 * A_{77} \leq 1$$

$$0 \leq Z'_{70} + U_8 - 2 * A_{78} \leq 1$$

$$0 \leq Z'_{70} + U_9 - 2 * A_{79} \leq 1$$

$$0 \leq Z'_{70} + U_{10} - 2 * A_{710} \leq 1$$

$$\begin{aligned}
L_1(A_{72} + A_{73} + A_{74} + A_{75} + A_{76} + A_{77} + A_{78} + A_{79} + A_{710}) &\leq y_{71} \\
&\leq L_1(A_{71} + A_{72} + A_{73} + A_{74} + A_{75} + A_{76} + A_{77} + A_{78} + A_{79} + A_{710})
\end{aligned}$$

$$\begin{aligned}
L_2(A_{73} + A_{74} + A_{75} + A_{76} + A_{77} + A_{78} + A_{79} + A_{710}) &\leq y_{72} \\
&\leq L_2(A_{72} + A_{73} + A_{74} + A_{75} + A_{76} + A_{77} + A_{78} + A_{79} + A_{710})
\end{aligned}$$

$$\begin{aligned} L_3(A_{74} + A_{75} + A_{76} + A_{77} + A_{78} + A_{79} + A_{710}) &\leq y_{73} \\ &\leq L_3(A_{73} + A_{74} + A_{75} + A_{76} + A_{77} + A_{78} + A_{79} + A_{710}) \end{aligned}$$

$$\begin{aligned} L_4(A_{75} + A_{76} + A_{77} + A_{78} + A_{79} + A_{710}) &\leq y_{74} \\ &\leq L_4(A_{74} + A_{75} + A_{76} + A_{77} + A_{78} + A_{79} + A_{710}) \end{aligned}$$

$$L_5(A_{76} + A_{77} + A_{78} + A_{79} + A_{710}) \leq y_{75} \leq L_5(A_{75} + A_{76} + A_{77} + A_{78} + A_{79} + A_{710})$$

$$L_6(A_{77} + A_{78} + A_{79} + A_{710}) \leq y_{76} \leq L_6(A_{76} + A_{77} + A_{78} + A_{79} + A_{710})$$

$$L_6(A_{78} + A_{79} + A_{710}) \leq y_{77} \leq L_7(A_{77} + A_{78} + A_{79} + A_{710})$$

$$L_8(A_{79} + A_{710}) \leq y_{78} \leq L_8(A_{78} + A_{79} + A_{710})$$

$$L_9(A_{710}) \leq y_{79} \leq L_9(A_{79} + A_{710})$$

$$0 \leq y_{710} \leq L_{10}(A_{710})$$

$$0 \leq Z'_{80} + U_1 - 2 * A_{81} \leq 1$$

$$0 \leq Z'_{80} + U_2 - 2 * A_{82} \leq 1$$

$$0 \leq Z'_{80} + U_3 - 2 * A_{83} \leq 1$$

$$0 \leq Z'_{80} + U_4 - 2 * A_{84} \leq 1$$

$$0 \leq Z'_{80} + U_5 - 2 * A_{85} \leq 1$$

$$0 \leq Z'_{80} + U_6 - 2 * A_{86} \leq 1$$

$$0 \leq Z'_{80} + U_7 - 2 * A_{87} \leq 1$$

$$0 \leq Z'_{80} + U_8 - 2 * A_{88} \leq 1$$

$$0 \leq Z'_{80} + U_9 - 2 * A_{89} \leq 1$$

$$0 \leq Z'_{80} + U_{10} - 2 * A_{810} \leq 1$$

$$\begin{aligned} L_1(A_{82} + A_{83} + A_{84} + A_{85} + A_{86} + A_{87} + A_{88} + A_{89} + A_{810}) &\leq y_{81} \\ &\leq L_1(A_{81} + A_{82} + A_{83} + A_{84} + A_{85} + A_{86} + A_{87} + A_{88} + A_{89} + A_{810}) \end{aligned}$$

$$\begin{aligned}
L_2(A_{83} + A_{84} + A_{85} + A_{86} + A_{87} + A_{88} + A_{89} + A_{810}) &\leq y_{82} \\
&\leq L_2(A_{82} + A_{83} + A_{84} + A_{85} + A_{86} + A_{87} + A_{88} + A_{89} + A_{810}) \\
L_3(A_{84} + A_{85} + A_{86} + A_{87} + A_{88} + A_{89} + A_{810}) &\leq y_{83} \\
&\leq L_3(A_{83} + A_{84} + A_{85} + A_{86} + A_{87} + A_{88} + A_{89} + A_{810}) \\
L_4(A_{85} + A_{86} + A_{87} + A_{88} + A_{89} + A_{810}) &\leq y_{84} \\
&\leq L_4(A_{84} + A_{85} + A_{86} + A_{87} + A_{88} + A_{89} + A_{810}) \\
L_5(A_{86} + A_{87} + A_{88} + A_{89} + A_{810}) &\leq y_{85} \leq L_5(A_{85} + A_{86} + A_{87} + A_{88} + A_{89} + A_{810}) \\
L_6(A_{87} + A_{88} + A_{89} + A_{810}) &\leq y_{86} \leq L_6(A_{86} + A_{87} + A_{88} + A_{89} + A_{810}) \\
L_6(A_{88} + A_{89} + A_{810}) &\leq y_{87} \leq L_7(A_{87} + A_{88} + A_{89} + A_{810}) \\
L_8(A_{89} + A_{810}) &\leq y_{88} \leq L_8(A_{88} + A_{89} + A_{810}) \\
L_9(A_{810}) &\leq y_{89} \leq L_9(A_{89} + A_{810}) \\
0 &\leq y_{810} \leq L_{10}(A_{810})
\end{aligned}$$

$$\begin{aligned}
0 &\leq Z'_{90} + U_1 - 2 * A_{91} \leq 1 \\
0 &\leq Z'_{90} + U_2 - 2 * A_{92} \leq 1 \\
0 &\leq Z'_{90} + U_3 - 2 * A_{93} \leq 1 \\
0 &\leq Z'_{90} + U_4 - 2 * A_{94} \leq 1 \\
0 &\leq Z'_{90} + U_5 - 2 * A_{95} \leq 1 \\
0 &\leq Z'_{90} + U_6 - 2 * A_{96} \leq 1 \\
0 &\leq Z'_{90} + U_7 - 2 * A_{97} \leq 1 \\
0 &\leq Z'_{90} + U_8 - 2 * A_{98} \leq 1 \\
0 &\leq Z'_{90} + U_9 - 2 * A_{99} \leq 1 \\
0 &\leq Z'_{90} + U_{10} - 2 * A_{910} \leq 1
\end{aligned}$$

$$\begin{aligned}
L_1(A_{92} + A_{93} + A_{94} + A_{95} + A_{96} + A_{97} + A_{98} + A_{99} + A_{910}) &\leq y_{91} \\
&\leq L_1(A_{91} + A_{92} + A_{93} + A_{94} + A_{95} + A_{96} + A_{97} + A_{98} + A_{99} + A_{910}) \\
L_2(A_{93} + A_{94} + A_{95} + A_{96} + A_{97} + A_{98} + A_{99} + A_{910}) &\leq y_{92} \\
&\leq L_2(A_{92} + A_{93} + A_{94} + A_{95} + A_{96} + A_{97} + A_{98} + A_{99} + A_{910}) \\
L_3(A_{94} + A_{95} + A_{96} + A_{97} + A_{98} + A_{99} + A_{910}) &\leq y_{93} \\
&\leq L_3(A_{93} + A_{94} + A_{95} + A_{96} + A_{97} + A_{98} + A_{99} + A_{910}) \\
L_4(A_{95} + A_{96} + A_{97} + A_{98} + A_{99} + A_{910}) &\leq y_{94} \leq L_4(A_{94} + A_{95} + A_{96} + A_{97} + A_{98} + A_{99} + A_{910}) \\
L_5(A_{96} + A_{97} + A_{98} + A_{99} + A_{910}) &\leq y_{95} \leq L_5(A_{95} + A_{96} + A_{97} + A_{98} + A_{99} + A_{910}) \\
L_6(A_{97} + A_{98} + A_{99} + A_{910}) &\leq y_{96} \leq L_6(A_{96} + A_{97} + A_{98} + A_{99} + A_{910}) \\
L_6(A_{98} + A_{99} + A_{910}) &\leq y_{97} \leq L_7(A_{97} + A_{98} + A_{99} + A_{910}) \\
L_8(A_{99} + A_{910}) &\leq y_{98} \leq L_8(A_{98} + A_{99} + A_{910}) \\
L_9(A_{910}) &\leq y_{99} \leq L_9(A_{99} + A_{910}) \\
0 &\leq y_{910} \leq L_{10}(A_{910})
\end{aligned}$$

$$\begin{aligned}
0 &\leq Z_{90} + U_1 - 2 * A_{101} \leq 1 \\
0 &\leq Z_{90} + U_2 - 2 * A_{102} \leq 1 \\
0 &\leq Z_{90} + U_3 - 2 * A_{103} \leq 1 \\
0 &\leq Z_{90} + U_4 - 2 * A_{104} \leq 1 \\
0 &\leq Z_{90} + U_5 - 2 * A_{105} \leq 1 \\
0 &\leq Z_{90} + U_6 - 2 * A_{106} \leq 1 \\
0 &\leq Z_{90} + U_7 - 2 * A_{107} \leq 1 \\
0 &\leq Z_{90} + U_8 - 2 * A_{108} \leq 1 \\
0 &\leq Z_{90} + U_9 - 2 * A_{109} \leq 1 \\
0 &\leq Z_{90} + U_{10} - 2 * A_{1010} \leq 1
\end{aligned}$$

$$\begin{aligned}
L_1(A_{102} + A_{103} + A_{104} + A_{105} + A_{106} + A_{107} + A_{108} + A_{109} + A_{1010}) &\leq y_{101} \\
&\leq L_1(A_{101} + A_{102} + A_{103} + A_{104} + A_{105} + A_{106} + A_{107} + A_{108} + A_{109} + A_{1010}) \\
L_2(A_{103} + A_{104} + A_{105} + A_{106} + A_{107} + A_{108} + A_{109} + A_{1010}) &\leq y_{102} \\
&\leq L_2(A_{102} + A_{103} + A_{104} + A_{105} + A_{106} + A_{107} + A_{108} + A_{109} + A_{1010}) \\
L_3(A_{104} + A_{105} + A_{106} + A_{107} + A_{108} + A_{109} + A_{1010}) &\leq y_{103} \\
&\leq L_3(A_{103} + A_{104} + A_{105} + A_{106} + A_{107} + A_{108} + A_{109} + A_{1010}) \\
L_4(A_{105} + A_{106} + A_{107} + A_{108} + A_{109} + A_{1010}) &\leq y_{104} \\
&\leq L_4(A_{104} + A_{105} + A_{106} + A_{107} + A_{108} + A_{109} + A_{1010}) \\
L_5(A_{106} + A_{107} + A_{108} + A_{109} + A_{1010}) &\leq y_{105} \leq L_5(A_{105} + A_{106} + A_{107} + A_{108} + A_{109} + A_{1010}) \\
L_6(A_{107} + A_{108} + A_{109} + A_{1010}) &\leq y_{106} \leq L_6(A_{106} + A_{107} + A_{108} + A_{109} + A_{1010}) \\
L_7(A_{108} + A_{109} + A_{1010}) &\leq y_{107} \leq L_7(A_{107} + A_{108} + A_{109} + A_{1010}) \\
L_8(A_{109} + A_{1010}) &\leq y_{108} \leq L_8(A_{108} + A_{109} + A_{1010}) \\
L_9(A_{1010}) &\leq y_{109} \leq L_9(A_{109} + A_{1010}) \\
0 &\leq y_{1010} \leq L_{10}(A_{1010})
\end{aligned}$$